GROUP PROJECT

COMP 2147

Kay Brooks
Rachel Bryant 100827445

GROUP#
CRN: 11571

ABC Clinic

System Service Request

REQUESTED BY John Doe DATE October 1, 2026

DEPARTMENT Administration / Clinic Operations

LOCATION Greater Toronto Area (GTA) Branch

CONTACT John.Doe@abcclinic.ca | (416) 555-0199

TYPE OF REQUEST

- [x] New System
- [ ] System Enhancement
- [ ] System Error Correction

URGENCY

- [ ] Immediate - Operations are impaired or opportunity lost
- [x] Problems exist, can be worked around
- [ ] Business losses can be tolerated until system installed

PROBLEM STATEMENT

ABC Clinic is opening a new family medicine and walk-in facility in the GTA, expecting to serve 1,500–2,000 patients weekly with a staff of 10 physicians, 6 nurses, 8 receptionists, and 2 assistant managers. Government regulations mandate that patient medical records and personal information be stored and safeguarded electronically to comply with strict confidentiality and privacy laws. Currently, the clinic lacks an integrated electronic system to manage appointments, walk-in queues, triage prioritization, patient visit histories, lab requisitions, specialist referrals, and electronic billing (Ministry of Health and uninsured services). Without an automated Clinical Management System, the clinic faces operational bottlenecks, potential privacy compliance risks, manual paper tracking inefficiencies, and delayed patient care.

SERVICE REQUEST

I request the analysis, design, and development of a comprehensive, paperless Clinical Management System (CMS). The system should manage online and phone appointment bookings, walk-in queues and triage tagging, real-time health card verification with the Ministry, electronic health record (EHR) entries by nurses and physicians, automated electronic/printed prescriptions, lab requisitions and test result tracking, specialist referrals, Accounts Receivable for non-covered cash/insurance payments, and staff shift scheduling/payroll integration. The total allocated budget is $1,000,000, with a target implementation timeline of 9 months.

IS LIAISON Lead Systems Analyst / Project Lead

IS SPONSOR John Doe, Clinic Manager

TO BE COMPLETED BY PRIORITY BOARD

- [x] Request approved
- [ ] Recommend revision
- [ ] Recommend user development
- [ ] Reject for reason

Assigned to Systems Development Team A

Start date November 1, 2026

ABC Clinic **Prepared: October 1, 2026**

Project Charter

Project Name: Clinical Management System (CMS)

Project Manager: Kay Brooks (kbrooks@abcclinic.ca) & Rachel Bryant (rbryant@abcclinic.ca)

Customer: ABC Clinic Operations & Medical Staff

Project Sponsor: John Doe, Clinic Manager (jdoe@abcclinic.ca)

Project Start/End (projected): November 1, 2026 – July 31, 2027 (9 Months)

Project Overview:

This project will develop and deploy a comprehensive, paperless **Clinical Management System (CMS)** for the new ABC family medicine and walk-in clinic in the Greater Toronto Area. The purpose of this system is to streamline clinic workflows serving 1,500–2,000 patients weekly, automate patient scheduling and walk-in queue management with triage prioritization, secure electronic medical records in compliance with government privacy regulations, enable electronic billing (Ministry of Health and Accounts Receivable for non-covered/insurance services), and facilitate electronic exchanges for prescriptions, lab requisitions, and specialist referrals.

Objectives:

- **Regulatory Compliance:** Securely store and safeguard patient electronic health records in strict compliance with governmental privacy and confidentiality mandates.
- **Workflow & Operations Automation:** Automate appointment booking, walk-in queueing, triage jumping, and status tracking (booked, arrived, checked in, checked out, LWT, no-show).
- **Real-time Verification:** Provide real-time Health Card verification directly linked with the Ministry of Health system upon patient check-in.
- **Electronic Diagnostics & Referrals:** Enable doctors to generate and transmit electronic/printed prescriptions, lab requisitions (and receive lab results), and specialist referrals.
- **Financial & Payroll Support:** Support direct government billing for physicians, manage Accounts Receivable/Payable for non-covered cash/insurance items, and track non-salaried staff working hours for payroll processing.
- **Timely Delivery & Budget Adherence:** Complete full system deployment within the $1,000,000 budget and the 9-month timeframe.

Key Assumptions:

- The budget limit is fixed at $1,000,000 and the project timeline cannot exceed 9 months.
- External entities (Ministry of Health, third-party medical labs, pharmacies, specialist offices, and accounting firm) will provide compatible electronic interfaces or APIs.
- Staffing levels will remain approximately 10 physicians, 6 nurses, 8 receptionists, and 2 assistant managers.
- System interfaces will be web-browser/cloud-based for accessibility across examination rooms and front desk stations.
- Paper records received (such as outside lab results or specialist notes) can be scanned and securely stored in the system database.

Stakeholders and Responsibilities:

Stakeholder	Role	Responsibility	Signatures
John Doe	Clinic Manager / Project Sponsor	Project Vision, Budget Approval ($1M), Scope Sign-off, Operations Alignment	*John Doe*
Kay Brooks	Co-Project Manager	Project Planning, Monitoring, Execution, Resource Management, Deliverables	*Kay Brooks*
Rachel Bryant	Co-Project Manager	Requirements Architecture, Testing Strategy, Risk Management, Deliverables	*Rachel Bryant*
Lead Physician	User Representative (Doctors)	Clinical Workflow Functionality, EHR Specs, E-Prescriptions & Billing Needs	*Lead Physician*
Lead Nurse	User Representative (Nurses)	Vitals/Symptom Entry Functionality & Triage Workflow Requirements	*Lead Nurse*
Head Receptionist	User Representative (Front Desk)	Scheduling, Walk-in Queue, Health Card Verification & Payment Forms	*Head Receptionist*

ABC Clinic

Date: October 26, 2026

Project Scope Statement

Prepared by:

General Project Information

Project Name: Clinical Management System (CMS)

Sponsor: John Doe, Clinic Manager (jdoe@abcclinic.ca)

Project Manager: Kay Brooks (kbrooks@abcclinic.ca) & Rachel Bryant (rbryant@abcclinic.ca) Manager

Problem/Opportunity Statement:

ABC Clinic sees around 1,500 to 2,000 patients every week. Managing all these visits on paper makes check-ins slow, creates administrative headaches, and risks non-compliance with health privacy rules. The clinic needs an easy-to-use, paperless system to handle patient care smoothly and keep data secure.

Project Objectives:

- Build and launch the system within 9 months on a $1 million budget.
- Set up real-time health card checks with the Ministry of Health and securely store patient histories, lab results, and specialist notes.
- Speed up front-desk check-ins, manage walk-in queues, and handle e-prescriptions and payments.
- Help managers easily create shift schedules and track working hours for hourly staff.

Project Description:

We will build a complete software system for ABC Clinic. It will handle online and phone appointment bookings, 24-hour cancellation rules, walk-in triage queues, digital patient records, prescriptions, lab and specialist notes, patient payments for non-covered services, and staff scheduling.

Business Benefits:

Follows Privacy Rules: Keeps patient data safe and meets government privacy laws.
Faster Front Desk: Helps patients check in quicker and keeps the wait line moving smoothly.
Better Care: Gives doctors instant access to medical histories, test results, and notes.
Easier Money Tracking: Makes it simple to bill for paid services and track staff work hours.

Project Deliverables:

System Request & Project Charter
Feasibility Report & Financial Spreadsheet (NPV, ROI, Payback)
Scope Statement & Intangible Benefits List
System Diagrams (Use Cases, Workflows, Data Flow)
Database Models (ERD, Physical Model, Class Diagram)
Screen Mockups & Flowcharts
Testing Plan & Final Report

Estimated Project Duration:

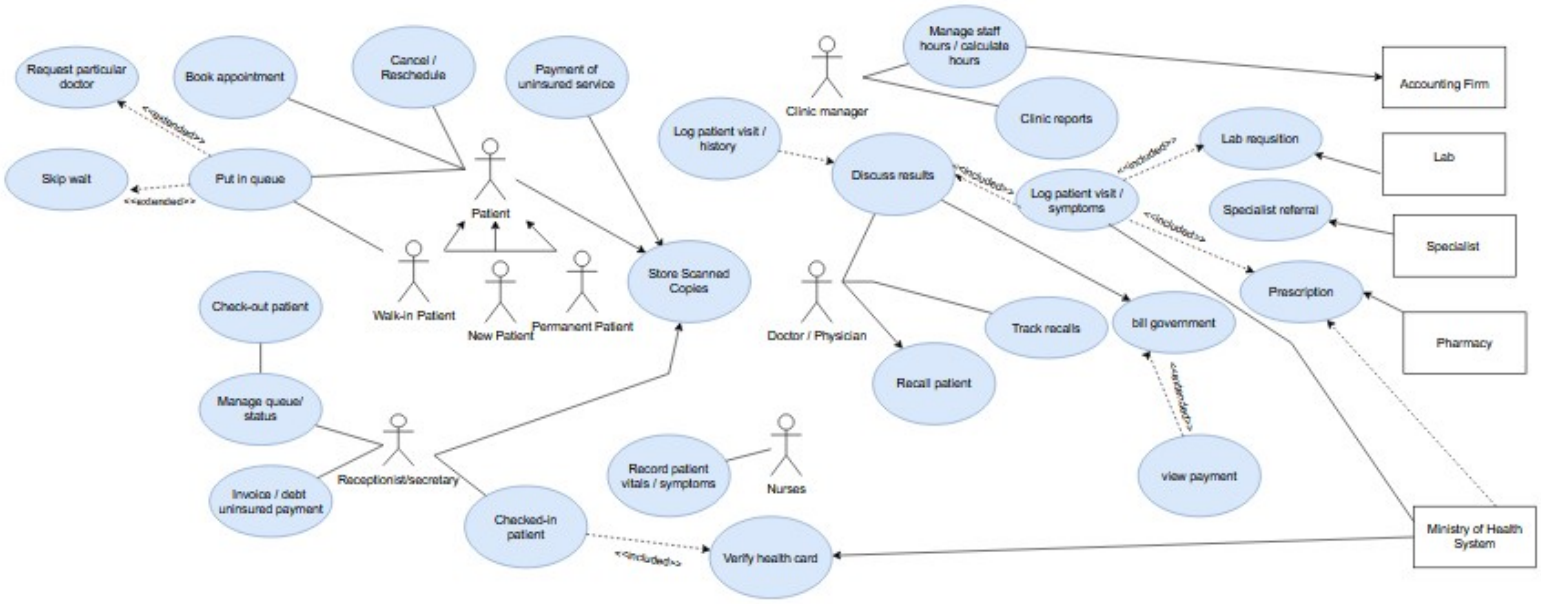
9 months

Section 10: Refined and Detailed Use Case Diagram

Project: Clinical Management System (CMS) — ABC Clinic

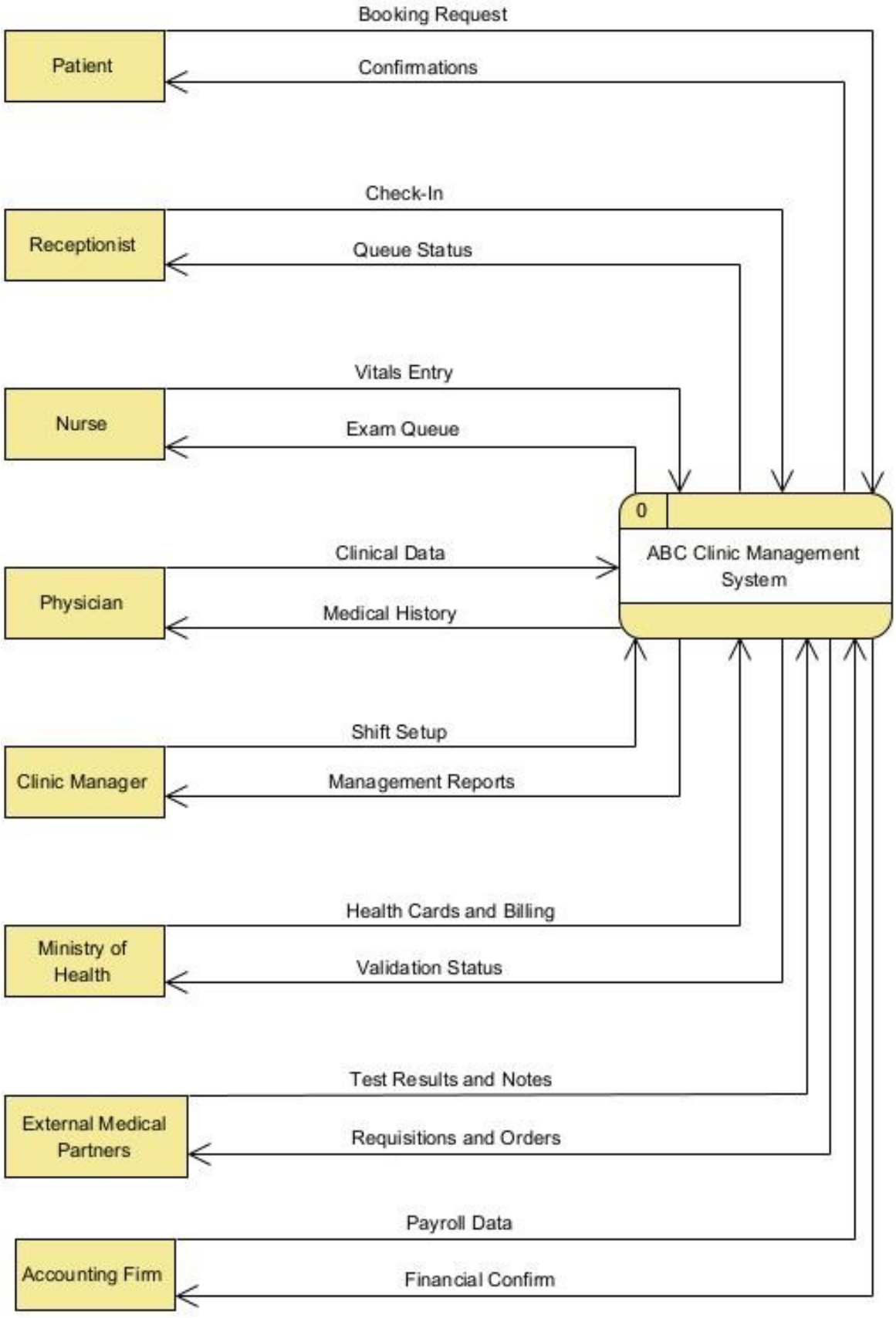
Description:

This refined and detailed use case diagram builds upon the initial textual analysis to incorporate complete actor generalization hierarchies (such as Patient and Staff subtypes), primary system actors (Patients, Receptionists, Nurses, Physicians, Clinic Manager, Ministry of Health, Accounting Firm, and External Medical Partners), and advanced system relationships—including `<<include>>` and `<<extended>>` dependencies—derived from the clinic's core workflows.



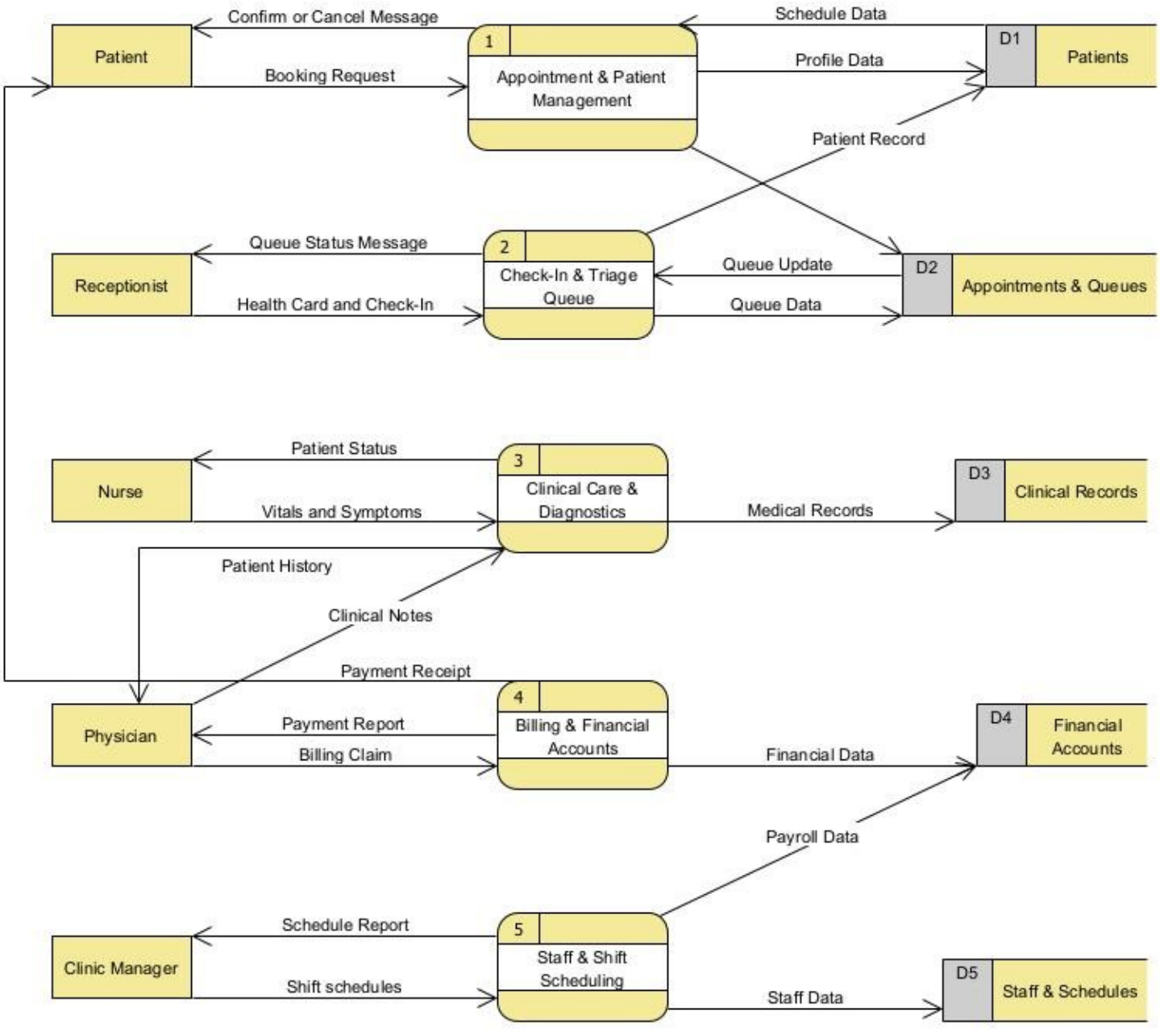
Visual Paradigm Standard (Arturo George Brown College)

Context-Level Data Flow Diagram (DFD) for the ABC Clinic Management System



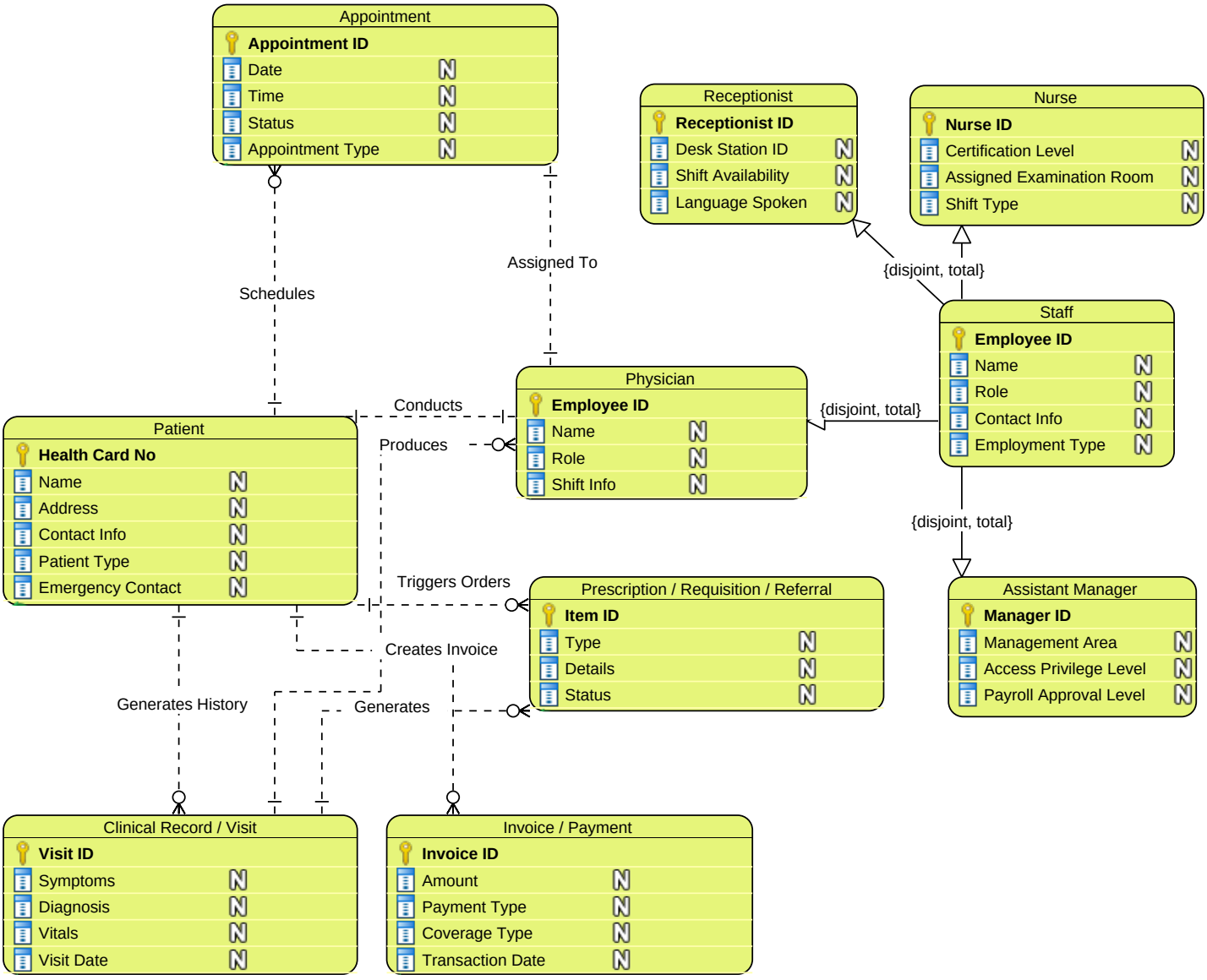
Visual Paradigm Standard (Admin(George Brown College))

Level-0 Data Flow Diagram



Visual Paradigm Standard(Admin(George Brown College))

ABC Clinic Management System - Conceptual ER Model (Entity-Relationship)



Visual Paradigm Standard(Admin(George Brown College))

ABC Clinic Management System - Physical Data Model (Relational Schema)

